Project Planning

Project: Rising Water – AI-Based Flood Prediction System

1. Project Overview

Rising Water is a Machine Learning-based flood prediction system developed to estimate flood probability using environmental parameters and provide an easy-to-use web interface for prediction.

2. Objectives

- Develop a flood prediction model.
- Build a Flask web application.
- Provide real-time prediction results.
- Support disaster preparedness.

3. Work Breakdown Structure (WBS)

Task	Description
Requirement Analysis	Understand problem and collect dataset
Data Preprocessing	Clean and prepare dataset
Model Development	Train XGBoost model
Web Development	Develop Flask application
Testing	Validate predictions and UI
Documentation	Prepare project documents
Deployment	Upload to GitHub and deploy

4. Project Timeline

Phase	Duration	Status
Planning	Week 1	Completed
Data Preparation	Week 1	Completed
Model Building	Week 2	Completed
Flask Development	Week 2	Completed
Testing	Week 3	Completed
Documentation	Week 3	In Progress

5. Resources

- Python
- Flask
- Pandas & NumPy
- Scikit-learn & XGBoost
- VS Code
- GitHub
- Kaggle Flood Dataset

6. Risks & Mitigation

Risk	Mitigation
Poor data quality	Data cleaning and validation
Model accuracy	Hyperparameter tuning
Deployment issues	Local testing before deployment

7. Deliverables

- Trained ML model
- Flask web application
- Project documentation
- GitHub repository

8. Conclusion

The project planning phase defines the activities, timeline, resources, and deliverables required to successfully complete the Rising Water project.